Frequently Asked Questions

Cosmic Calculation Series · Teaching logistics

WHAT IS THE COSMIC CALCULATION SERIES?

A collection of astronomy learning materials organized around the IOAA syllabus, intended to support physical understanding and curiosity about the universe.

WHAT IS AVAILABLE FOR BASIC ASTROPHYSICS?

Each of the five Basic Astrophysics topics has an HTML lecture, 15 original problems, fully worked solutions, vector figures and an interactive experiment, with downloadable PDFs. The earlier theory PDFs are retained as archives. Original club practice exercises are available for the two Coordinates and Times topics.

ARE SOLUTIONS AVAILABLE?

Basic Astrophysics (01.01–01.05) has 75 fully worked solutions, and Cosmology (06.01) has 30. These are independent MAIS PAC solutions to original exercises, not official marking schemes. Other curriculum solution PDFs remain placeholders; the introductory club handouts already have their own solutions.

WHAT BACKGROUND DOES THE SYLLABUS REQUIRE?

The IOAA syllabus requires high-school physics and mathematics. Standard solutions should not require complex numbers or solving differential equations. The listed mathematical methods include basic calculus.

WHAT DOES (Q) MEAN?

Qualitative understanding only. Quantitative reasoning or proficiency in topics marked (Q) is not mandatory.

IS THIS AN OFFICIAL IOAA PUBLICATION?

No. This is a MAIS PAC study resource. The syllabus PDF is a reformatted study edition of the syllabus; the topic codes and practical-topic labels are club organizational aids.

WHERE ARE THE EXISTING CLUB HANDOUTS?

The introductory physics and astronomy sheets, their solutions, and downloadable PDF versions are linked in the Teaching page's Existing club handouts section.